
2023**High School Mathematical Contest in Modeling (HiMCM) & Middle Mathematical Contest in Modeling (MidMCM)****Team Control Number: 14900****Problem Chosen: B**

Electric vehicles are now the leading form of environmental protection, but the government has a major problem. There is a gap in the gap of private and public: They can't control the number of private vehicles, and electric buses might just be the answer. As a public vehicle, electric buses (or e-buses) are one of the most effective methods with large travel time and existing manufacturers. Our model is designed to solve the financial, societal, and ecological issues that governments and bus drivers have in transitioning to an electric-dominated bus fleet, by computing the costs utilizing leading factors and comparing it to diesel buses (which sometimes are referred to as regular buses in this report).

The models created use many variables to inform us of the amount of e-buses required to be made in order for their purpose to be sufficiently fulfilled, taking into account their ecological benefits, financial costs, time duration for the implementation, and more.

The price being minimized by fewer e-buses is the immediate cost of constructing each raw e-bus. The saved money stems from the constant factors, costs that come during the working phase of a diesel bus and an electric bus, and are assumed to stay the same throughout. The three main constant assumptions are the ecological profits (saved money) which are from gas emission prevention and vehicle maintenance. Several factors have been considered as negligible in order to simplify this problem and use the importance of the factors considered. These models work side-by-side and aim to give a way to predict the effects of a gradual change to an e-bus fleet.

We will be demonstrating the model by applying it to a real-world country and taking information from that country then simulating the process. Australia is our primary example since it has clear intentions to transition to a fully electric bus fleet, but is only in its beginning phase where planning and visualization is needed to work from the current small percentage of electric buses. Therefore, a working roadmap was developed that instead of computing how to maximize our budget for the long-term ecological profits, it simulates the effects if we fit the entire replacement of every bus to electric in a 10-year timeslot.

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1 | Introduction

Buses are the second most traveled transportation in terms of kilometers traveled per person, only behind private vehicles (European Commission, 2023). Unfortunately, not only does this make buses a leading form of long-distance travel, but also a leading form of air pollution, greenhouse gas build up, and climate change. Over 2.4 million out of the approximate 3 million buses on the globe run on fuels such as diesel, compressed natural gas, and petroleum (Sustainable Bus, 2019).

The solution is the alternative — e-buses.

Countries around the world have begun their transition away from these fuel-powered buses to electrical buses as the benefits that these electrical buses bring may prove to later be more sustainable as a mode of transportation both economically and ecologically. Once implemented, these electronic buses aim to solve the environmental problems caused by diesel-powered buses, while also promising to have a lower maintenance cost, thus being more cost effective in the long run.

As traditional transportation continues its crucial purpose for convenient mobility, its side effects through the processing of the resources it utilizes ended up becoming a leading cause for many issues. These issues include climate change due to the greenhouse gas build up for which the transport sectors are consistently accountable for more than 20% of it. Additionally, traditional buses produce 1.3kg of CO₂ for every kilometer traveled which contributes to air pollution in stark contrast to electric buses which have zero emissions. That is why the complete transition to fleets of electric buses can greatly improve our carbon footprint (Prytz, 2022).

However, the implementation of introducing these new electric buses brings many financial complications especially as it is an upfront investment for the future of transportation. Examples of such complications are the costs that need to be covered for the electric buses themselves and their charging stations or pantographs (an established powerline used to charge electronic buses on the move) which together already takes up hundreds of thousands of dollars for each set of electronic buses, and with the number of electric buses needed to phase out traditional buses, it can take up even millions of dollars upfront.

What makes electronic buses an investment is that the total cost of ownership of these electric buses will gradually decrease as the majority of its expenses fall under capital cost, including batteries and charging parts, which will be improved over time. Compared to the current expenses of diesel or petrol-powered buses, the costs to own and maintain electric buses are projected to become lower and more stable as time progresses, also taking into account that with the minimal parts of electric buses, their costs are more stable and less prone to fluctuation (Clean Mobility Shift, 2021).

With all of this in mind, Australia is a country attempting to fully transition into electronic buses, and we have constructed models in order to predict the consequences of such a transition on the country ecologically and financially.

2 | Model and Solution

These two models attempt to predict the ecological consequences of a full transition to e-buses and the financial implications of a gradual transition to e-buses, respectively. The way we have used the information of metropolitan areas is by converting all the information into monetary values, as will be evident later on. These models are a way of optimizing the price of transition to e-buses, while still being able to impact the environment positively.

4.1 Variables and Definitions

Variable	Definition
c_{e-bus}	The cost of one e-bus alone.
$c_{charging\ station}$	The cost of one charging station that is capable of charging two e-buses completely every 24 hours.
$c_{maintenance}$	The monthly cost of maintaining an e-bus (repairs, electricity).
α	The number of e-buses to be made. This value is unknown.
$\alpha_{monthly}$	The number of e-buses made monthly. This will be given by α .
w	The wage of a regular bus driver.
c_{fare}	The fare price of one passenger to ride on a regular bus.
n	The average number of passengers one regular bus would receive per month.
m	Number of months it takes to construct and implement one e-bus.
$c_{societal}$	The “societal cost” that comes with a transition of one e-bus.
$c_{transition}$	The cost of transition of 1 e-bus.
$c_{monthly}$	The initial cost of transition of e-buses plus the monthly cost of maintaining e-buses.
f_{total}	The total funds available to the government for a full transition to e-buses.

C_{total}	The total cost of transition to e-buses.
$C_{gas\ emissions}$	The annual cost spent due to gas emissions from our diesel/gas buses.
$C_{public\ health}$	The annual cost spent on public health due to diesel/gas buses and climate change.
$C_{noise\ pollution}$	The annual loss of the government due to noise pollution.
α_{buses}	Total number of registered buses.
$\alpha_{initial}$	Number of initial diesel buses before the transition.
α_{diesel}	Number of diesel buses. This value is unknown.
C_{saved}	The amount of money saved per month due to the transition to e-buses.
C_{lost}	The amount of money lost per month due to the transition to e-buses.
$C_{ecologic}$	The total profit <i>only</i> from ecological impacts per month.
C_{profit}	The profit per month of transitioning to an all e-bus fleet taking amount of money saved with e-buses and total cost of transition to e-buses into consideration.
t	The time taken for the transition (in months).
i	The number of months since the beginning of the transition.
M	Number of e-buses who have begun manufacturing.
I	Number of e-bus with implementation.
y	The actual year.
r	Function showing number of buses manufactured from the start to the given year.
c	Function showing number of buses implemented from start to the given year.

4.2 Ecological Consequences

4.2.1 Overview

This model will analyze and exhibit both the positive and negative ecological consequences that will occur by transitioning to an all e-bus fleet. Here, the goal is to balance both the positive and negative ecological consequences of transitioning to e-buses such that the positive ecological consequences outweigh the negative consequences. To do this, the costs of both types of consequences - the 'gains' (positive consequences) and the 'losses' (negative consequences) - will be listed together so that these can be put together and analyzed.

By analyzing the gains and losses, an understanding of the meaning of transitioning to an all e-bus fleet is gained. First, this transition shows three (3) main benefits all related to climate change. These are (1) Lessened Gas Emissions (from air pollution and greenhouse gases), (2) Improved Public Health, and (3) Reduced Noise Pollution. These positive consequences bring more comfort, sustainability, and practicality to the societies which in turn reduces the cost required in these ecological scenarios. On the other hand, there are negative consequences which relate to the financial aspect of these e-buses. Though these e-buses will definitely help cities save money due to the lessened ecological effects with the transition, a large amount of money will also be required to purchase, transition, and use e-buses in our cities. This will be discussed thoroughly in our second model. It is important to note however that comparing the money saved by transitioning and money spent by transitioning is an important step to show that the positives exceed the negatives even when the transition is ongoing. The positives will already clearly exceed the negatives once the transition is over, because there will be no money spent on the requirements of e-buses.

4.2.2 The Model

The assumption made here is that the amount of money saved by the government with these e-buses would be equal through all months. It is important to make this assumption because the data we have about these values are by year. By making this assumption, we can easily compare all variables by the month. Doing this allows us to understand how the positive ecological consequences will outweigh the negative. Another assumption is that once e-buses were to be implemented completely in a certain area, money would no longer be spent in that area on reducing gas emissions, treatment of diseases that are due to harmful gasses, and reducing noise pollution.

By this,

$$c_{monthly} < c_{ecologic}$$

$$c_{lost} < c_{saved}$$

To solve,

$$c_{profit} = c_{ecologic} - c_{monthly}$$

$$\begin{aligned}
c_{\text{saved}} &= \alpha \left(\frac{(c_{\text{gas emissions}} + c_{\text{public health}} + c_{\text{noise pollution}})}{12 \times \alpha_{\text{initial}}} \right) \\
c_{\text{lost}} &= \alpha_{\text{diesel}} \left(\frac{(c_{\text{gas emissions}} + c_{\text{public health}} + c_{\text{noise pollution}})}{12 \times \alpha_{\text{initial}}} \right) \\
\alpha_{\text{diesel}} &= \alpha_{\text{buses}} - \alpha \\
c_{\text{ecologic}} &= c_{\text{saved}} - c_{\text{lost}}
\end{aligned}$$

The constraint in this model essentially states that the monthly transition cost has to be less than the monthly amount saved on ecological efforts. This will require the profit to be positive. This is because if it were the other way around, this would prove that the transition and implementation of electric buses is an ineffective solution. Here, it is important that we also get a positive profit. Negative profits mean that money is lost.

The monthly profit is given by the amount of money saved per month (with the reduction of these ecological problems) minus the cost of transition of e-buses per month. Here, the amount of money saved each month is solved by multiplying the number of e-buses to be made by the money spent annually on our three main issues—gas emissions, public health, and noise pollution - that is then divided by 12 and the initial number of regular buses for it to be by month. The number of e-buses made is multiplied to what is saved to know how at least how many e-buses are required to profit. To put it simply, this part of the model wants to minimize the number of e-buses. To get the money lost on ecological efforts, a similar model is applied but the number of buses to be made is replaced by the number of diesel buses currently existing (this number is also unknown and relies on α). To get the number of diesel buses, we simply subtract the number of e-buses to be made from the total number of buses in a given area. The money saved ecologically is naturally given by the amount gained (or saved) subtracted by the amount lost. The money saved is the money we get to use in other areas after a full e-bus transition because the issues are now reduced making more funds available.

4.2.3 Application and Analysis

This model will be applied to Australia. That being said, all the price values will be in Australian dollars (AUD). It is important to note that as of 2023, Australia has a 0.2% electric bus fleet, and they are currently in the transition process to a full (or majority) e-bus fleet.



Image 1: An e-bus in Australia (photo taken by one of our members).

To apply this model, the values of the known variables should first be identified. In this application, the value of c_{saved} , c_{loss} and $c_{ecologic}$ from the values of $c_{gas\ emissions}$, $c_{public\ health}$, and $c_{noise\ pollution}$ will be determined. The values of c_{profit} , $c_{monthly}$, and c_{total} will be solved in the second model that will be applied in the later sections.

First, the cost of gas emissions is calculated by:

For diesel vehicles: 166 AUD per kg of diesel for health costs, and 70 AUD per kg of diesel for gas emissions. 70 divided by 166 = 42.17% of health costs (42.17% of 264 million AUD), which results in ~111 million AUD spent on gas emissions.

Therefore,

$$c_{gas\ emissions} = 111,328,800$$

Next, the cost of public health is calculated by:

Approximately 3,420 AUD is spent on health per diesel vehicle, and there are 77.35 thousand registered buses of which 99.8% are diesel, so there are 77,195 diesel buses initially.

From this,

$$\begin{aligned}\alpha_{buses} &= 77,350 \\ \alpha_{initial} &= 77,195\end{aligned}$$

$$77,195 \times 3,420 \text{ AUD} = \text{Approximately } 264,000,000 \text{ AUD}$$

Therefore,

$$c_{public\ health} = 264,000,000 \text{ AUD}$$

The cost of noise pollution is calculated by:

Lastly, the annual cost of noise pollution per vehicle in Australia is 250 AUD. As mentioned in the previous part, we also have 77,195 diesel buses.

$$77,195 \times 250 \text{ AUD} = \text{Approximately } 19,300,000 \text{ AUD}$$

Therefore,

$$c_{noise\ pollution} = 19,300,000 \text{ AUD}$$

With all of these,

$$c_{saved} = \alpha \left(\frac{(c_{gas\ emissions} + c_{public\ health} + c_{noise\ pollution})}{12 \times \alpha_{initial}} \right) = \alpha \left(\frac{111,328,800 + 264,000,000 + 19,300,000}{12 \times 77,195} \right) = \text{Approximately } 1710\alpha$$

$$c_{saved} = 1710\alpha$$

Which implies,

$$c_{lost} = 1710\alpha_{diesel}$$

Which gives,

$$c_{ecological} = 1710\alpha - 1710\alpha_{diesel} = 1710(\alpha - \alpha_{diesel})$$

Applied to $\alpha_{diesel} = \alpha_{buses} - \alpha$,

$$c_{ecological} = 1710(\alpha - (77,350 - \alpha)) = 1710(2\alpha - 77350) = 3420\alpha - 132,268,500$$

This is the amount of money saved per month on ecological impacts if Australia were to convert to an electric bus fleet. Keep in mind that this does not factor in the monthly cost of transitioning to e-buses per month as the focus is only the amount that would be spent on ecological impacts per month. The profit per month has not been calculated yet. Only once the profit per month has been calculated will we be able to present the effectiveness of e-buses in our society.

4.3 Financial Implications

4.3.1 Overview

Our model will simulate the financial implications of gradually transitioning to an all e-bus fleet, but it will also show how many e-buses are needed to make “enough” of a positive impact ecologically. The two primary reasons for transitioning to e-buses are to reduce maintenance cost, and create a more sustainable and environmentally-healthy method for travel in order to fight against environmental crises such as climate change by reducing the gasses spread by regular buses. This model aims to optimize the cost of transitioning, while ensuring that the environment is affected positively.

4.3.2 The Model

The assumption here is that given infinite money, the government would automatically implement a complete e-bus fleet, and phase out all regular buses that run on gas. Since infinite money is definitely not included, the problem becomes how to minimize the cost as well as the time taken while maximizing the ecological benefit, which in the previous section was quantified by looking at how much money is saved monthly on environmental efforts by helping the environment early on. There are two constraints in this model, and two things to push to their extremes.

Minimize c_{total} and t , maximize $c_{ecological}$

Subject to

$$\begin{aligned} c_{monthly} &< c_{ecological} \\ c_{total} &< f_{total} \end{aligned}$$

Where

$$\begin{aligned} c_{total} &= \frac{\alpha(c_{transition})}{2} \\ t &= \frac{\alpha}{\alpha_{monthly}} \\ c_{monthly} &= (c_{maintenance})(\alpha_{monthly}) + \frac{(c_{transition})(\alpha_{monthly})}{2} \\ c_{ecological} &= 1710(\alpha - \alpha_{diesel}) \\ \alpha_{diesel} &= \alpha_{buses} - \alpha \\ c_{transition} &= c_{e-bus} + \frac{c_{charging\ station}}{3} + c_{societal} \\ c_{societal} &= w + c_{fare}nm \end{aligned}$$

The first constraint requires that the money spent on transitioning to e-buses per month (which we will assume to be constant once the transition begins) is less than the money that would be spent on ecological efforts such as public health and projects with the intent to reduce noise pollution or slow down climate change if there were no e-buses present in the area. This value represents the money that would be saved if the area were to fully convert to e-buses. This constraint defines what

will be considered “enough” ecological benefit. It also ensures that the profit (c_{profit} , which was defined in the last model) is positive. The next constraint is already a given. The total amount of cost for the full transition has to be less than or equal to the funds available for the transition.

The total cost is given by the number of e-buses to be made in total times the cost of transitioning for one e-bus, divided by 2, since external funding will cover 50% of the transition costs. The monthly cost for the transition depends on the monthly maintenance cost of one e-bus (repairs, check-ups) and the transition cost of the amount of e-buses made in that month divided by 2 (again, because external funding will cover 50% of the transition costs). For the maintenance cost, the cost of electricity during charging is negligible because the cost of gasses like diesel are more expensive than batteries.

The time taken for the transition is given by the amount of e-buses divided by the number of e-buses made per month. Both of these values are unknown as of now, and the solution is to find the highest amount of e-buses to be made per month while also keeping the price low.

The ecological cost or amount of money saved ecologically was determined by the last model and so was the amount of diesel buses. Since the number of diesel buses at a given time is unknown, it depends on the number of existing e-buses (or number of e-buses to be made, which is a changing value in the process).

The cost of transitioning one e-bus is determined by the average cost of one e-bus (which is assumed to be constant) plus the cost of a charging station divided by 3 (since based on research, a cheap high power level 2 charger can charge in at most 8 hours, so one charging station can provide for $24/8 = 3$ different e-buses, assuming they all make one trip per day that depletes all their charge) plus the societal cost of transitioning one e-bus.

The “societal cost” is considered, which gives monetary value to the amount lost by civilians. There will be regular bus drivers who will lose their jobs because of this transition to e-buses and passengers who won’t be able to travel because of the lack of buses while the e-buses are being constructed and implemented. The societal cost is decided by the average wage of a regular bus driver, added to the cost of fare for one person to ride the bus times the number of passengers the bus gets monthly times the number of months it takes to construct an e-bus and implement it (it can be assumed that once the e-bus is implemented, the bus driver will continue to drive that bus, so the societal cost is only the amount of income lost by the regular bus driver while their regular bus is phased out. It can also be assumed that the regular bus will be phased out the moment the construction on the e-bus begins.). The societal cost also assigns a quantitative value to the amount that passengers will lose.

The two unknowns here are α (the total number of e-buses to be made) and $\alpha_{monthly}$ (the total number of e-buses to be made per month). All the other information will be given by the metropolitan area this model is applied to.

4.2.4 Application and Analysis

This model will also be applied to Australia, so likewise, all the values will be in Australian dollars (AUD). Since Australia definitely does not have infinite money, the problem is to minimize the cost of a transition to e-buses, but not necessarily provide a way to fully transition to e-buses. All that is needed is to find out how many e-buses it takes to impact the environment positively.

Now, the given information will be identified.

The average cost of one e-bus is calculated by:

For this, the information will be taken from the Yutong company which manufactures and sells e-buses. A wide variety of electric buses have been chosen, which vary in both model, price, and length, so a more accurate average price can be found.

- Joylong E6 minibus - 140,000 AUD
- Yutong E10 bus - 480,000 AUD
- Yutong E12 bus - 528,000 AUD
- Electric school bus - 614,000 AUD
- Yutong C12E - 748,000 AUD

The average of this is 502,000 AUD.

Therefore,

$$c_{e-bus} = 502,000$$

The average cost of one charging station is calculated by:

A Type C ESB charger which is a high power level 2 charger requires a 6 to 8 hour charge. For the sake of simplicity, the charging time will be assumed to be the maximum of 8 hours. 3,000 to 8,000 AUD is spent on equipment, and approximately 15,500 AUD for installation. The average equipment cost is 5,500 AUD which when added to the installation cost, results in an average price of 21,000 AUD for a charging station which can provide for 3 e-buses in total.

Therefore,

$$c_{charging\ station} = 21,000$$

The average monthly cost of maintaining one e-bus is calculated by:

The economic cost lifespan of an e-bus is 727,055 AUD, and the lifespan of an e-bus is 12-15 years which is about 13.5 years. The cost of an e-bus is about 502,000 AUD.

727,055 AUD - 502,000 AUD = 225,000 AUD spent in lifetime on maintenance. 225,000 AUD divided by 162 months = Approximately 1,400 AUD per month on maintenance, which includes repairs and electricity.

Therefore,

$$c_{\text{maintenance}} = 1,400$$

The societal cost is given by:

$$c_{\text{societal}} = w + c_{\text{fare}}nm$$

Based on research,

$$w = 35.43 \text{ (AUD)}$$

$$c_{\text{fare}} = 4.71 \text{ (AUD)}$$

$$m = 9 \text{ (months)}$$

For both these values, the maximum cost is being used since the worst-case scenario needs to be considered for the model to be proven as effective.

The number of passengers per month is calculated by:

According to The Australian Bureau of Statistics, 673,969 households don't have a motor vehicle, primarily amongst lone households. This means a sensible assumption is for 673,969 people to commute every day. There are 77,195 registered diesel (regular) buses in Australia (as was calculated earlier).

For the sake of simplicity, this means every day a bus gets about $\frac{673,969 \times 2}{77,195 \div 2} = 35$ passengers. Every month, that's approximately $30 \times 35 = 1,050$.

Therefore,

$$n = 1,050$$

and

$$c_{\text{societal}} = w + c_{\text{fare}}nm = 35.43 + 4.71 \times 1050 \times 9 = 44,545 \text{ (AUD)}$$

$$c_{\text{societal}} = 44,545$$

The transition cost of one e-bus is given by:

$$c_{\text{transition}} = c_{\text{e-bus}} + \frac{c_{\text{charging station}}}{3} + c_{\text{societal}} = 502,000 + \frac{21,000}{3} + 44,545 = 553,545 \text{ (AUD)}$$

$$c_{\text{transition}} = 553,545$$

The monthly cost of transition to e-buses is calculated by:

$$c_{monthly} = (c_{maintenance})(\alpha_{monthly}) + \frac{(c_{transition})(\alpha_{monthly})}{2} = 278,172\alpha_{monthly}$$

The total cost of the transition is calculated by:

$$c_{total} = \frac{\alpha(c_{transition})}{2} = \frac{\alpha(553,545)}{2} = 276,772\alpha$$

The total funds for the transition, based on research, is 20 billion AUD.

$$f_{total} = 20,000,000,000$$

Now, the system of equations is

$$\begin{aligned} c_{total} &= 276,772\alpha \\ c_{ecological} &= 3420\alpha - 132,268,500 \\ c_{monthly} &= 278,172\alpha_{monthly} \end{aligned}$$

With the constraints

$$\begin{aligned} c_{monthly} &< c_{ecological} \\ c_{total} &< 20,000,000,000 \\ c_{ecological} &> 0 \end{aligned}$$

To apply this system, C++ code has been utilized in finding the most optimal values of α to maximize $c_{ecological}$ while minimizing c_{total} . $c_{ecological}$ breaks even and begins to profit for any $\alpha > 38,675$. This [code](#) was used to find it.

Based on this [code](#),

If

$$\alpha = 40,000$$

Then

$$c_{ecological} = 4,531,500$$

The most optimal value for $\alpha_{monthly}$ is 16, and it will cost 11 billion AUD over 2500 months which is approximately 208 years which is clearly not practical.

If

$$\alpha = 50,000$$

Then

$$c_{ecological} = 38,731,500$$

The most optimal value in this case for $\alpha_{monthly}$ is 139, and it will cost 14 billion AUD over 360 months, which is 30 years.

It can be seen that even an increase of 10,000 for α after breaking even can increase $c_{ecological}$ by 34,200,000 while also greatly reducing the time needed for the implementation. It would be a waste to use half the budget only to break even, so maximizing the use of the remaining budget to greatly increase profit and save time would be more preferable. Despite the rapidly increasing costs, the ecological profit ($\alpha_{ecological}$) and time saved is too great in comparison, thus it is most optimal to maximize the use of c_{total} in order to maximize $c_{ecological}$ in the least amount of time.

This [code](#) was used to find the value of α that would be most optimal for the maximum $c_{ecological}$ in the least amount of time within the given budget for the costs of c_{total} .

If

$$c_{total} = 20,000,000,000$$

Then

$$\begin{aligned}\alpha &= 72,261 \\ c_{ecological} &= 114864120\end{aligned}$$

It will take 175 months or 14.6 years to implement these e-buses with an optimal $\alpha_{monthly}$ of 412.

14.6 years is the fastest the e-buses will currently be implemented in Australia following the model, but more budgets, decreasing prices of materials to construct e-buses, and the money saved by easily maintaining these buses can be put back into model to make it more feasible for an implementation in under 10 years.

4.4 Roadmap

This 10-year roadmap, from 2024 to 2033, is an application of our two models above using our chosen area, Australia and later on, two other metropolitan areas namely Tokyo (Japan) and Seoul (South Korea), but instead of finding the minimum amount of e-buses, we are simulating a full transition to an e-bus fleet, while phasing out regular buses. In creating this, the goal is that all diesel buses (77,195 in this case) will be eradicated and replaced with e-buses by the year 2033. To do this, the basic plan of action is to gradually implement e-buses as transportation while also aiming for the maximum profit and the fastest implementation. Through this, the constraints mentioned above will be utilized to attain the realistic scenario of Australia and their e-bus fleet.

With these, our goal is to utilize the values in our model to create our roadmap. As mentioned multiple times in the creation of our models, the positives have to outweigh the negatives. This means even during the transition period, it is important that there can be a positive financial gain.

Here, it is important to note that the total number of time given to manufacture and implement all these e-buses in Australia is 120 months (from $10 \times 12 = 120$). As mentioned in the second model, a given e-bus takes 9 months at most to construct. This means that there are 111 months ($120 - 9 = 111$) to start manufacturing a group of e-buses. There will also be a constant number of e-buses to be made ($a_{monthly}$) so that the pace of implementation will stay on schedule and possible big losses will be avoided as losses can occur if too many e-buses are purchased in a given month.

With these in mind it is known that,

$$\frac{a}{a_{monthly}} = \frac{77,195}{a_{monthly}} = 111 \text{ (months)}$$

$$a_{monthly} = \frac{77,195}{111} = 696 \text{ (buses)}$$

It is known that there are about 77,195 diesel buses in Australia. With this, the goal by 2033 is that all the 77,195 diesel buses are replaced with e-buses. As mentioned before, the implementation of these e-buses will be constant all throughout the process. Because of this, the 77,195 e-buses will begin construction as groups in a span of 111 months. This means that each month, starting January 2024, 696 buses must begin construction that will last nine (9) months in total. A new group of 696 buses will begin its building from January 2024 to April 2033 so that by September 2024 to December 2033 there will be a new group of 696 e-buses per month to replace the existing diesel buses.

With this, it is known that the cost of transition per e-bus (including its actual cost, societal cost, and charging station), is about 553,545 AUD. Because of this, the total amount spent per month can be computed.

With this,

$$c_{monthly} = 553,545 \times 696 = 219,203,820 \text{ (AUD)}$$

This tells us that per month, the raw amount spent on the construction of new e-buses is 219,203,820 AUD.

However, the ecological costs must also be considered. For 111 months, starting when the e-buses are first implemented on the roads (September 2024), there will be gains because the assumption is that no money will be spent on these ecological consequences anymore due to these e-buses.

Applying our first model,

$$3,420(77,195) - 132,268,500 = 131,738,400$$

By this,

$$\frac{131,738,400}{111} = 1,186,833 \text{ (AUD)}$$

This tells us that per month, a total of 1,186,333 AUD is saved through the ecological consequences of transitioning to e-buses. This also means that:

$$219,203,820 - 1,186,833 = 218,016,987 \text{ (AUD)}$$

Here, we can see the total amount spent after removing the ecological cost that will be spent in the first place.

By this,

$$c_{total} = 218,016,987 \times 111 = \text{Approximately } 24,200,000,000 \text{ (AUD)}$$

As it can be seen, this cost goes above the total amount that the Australian government wants to place in the transition of e-buses by 4.2 billion AUD. However, the amount gained through the users of the implemented e-buses per month starting September 2024 should also be considered. To get this, an equation can be built.

$$c_{profit} = \sum_{i=1}^{i=111} a_{monthly} n c_{fare}^i$$

$$c_{profit} = \sum_{i=1}^{i=111} 696(1095)(2.70)i = \text{Approximately } 12,790,812,384 \text{ (AUD)}$$

In this equation, the process is to essentially multiply the number of buses that are made in a month by i (the number of months starting from September 2024), and it is also multiplied by the number of passengers and the fare of one bus ride. The values shown in the next parts were calculated through our previous models. With this model, the process is to get the number of e-buses during a given time and multiply it by the number of passengers and the fare. This is done 111 times because the duration is September 2024 to April 2033 or 111 months.

This equation essentially proves that through the transition to e-buses, a big benefit happens wherein the positive effects outweigh the negative effects. As shown by c_{total} , the amount that may have been spent all throughout the transition proves to be higher than the given budget for the decade. However, observing the profit during this time - considering the big amount of money pulled - about half the money (almost 12.8 billion AUD) was gained. This profit proves the financial potential of the e-buses in Australia especially when the transition is over.

By the end of it,

$$24,200,000,000 - 12,790,812,384 = 11,409,187,616 \text{ (AUD)}$$

This shows the total amount of money that will be spent by the Australian government when it comes to the transition to an all e-bus fleet. Here, it is important to note that this amount of 11.4 billion AUD is only about half of the total amount that the government placed into it for the full transition. This proves that the full e-bus fleet, together with our system, can work hand in hand in minimizing the amount required to transition to an e-bus fleet while allowing profit to generate.

Knowing this system and its strengths, the following schedule can be put on a table:

AUSTRALIA	Number of E-Buses Who Have Begun Manufacture Stage	Number of E-Buses With Implementation
by December 2024	8,352	2,784
December 2025	16,704	11,136
December 2026	25,056	19,488
December 2027	25,056	19,488
December 2028	41,760	36,192
December 2029	50,112	44,544
December 2030	58,464	52,896
December 2031	66,816	61,248
December 2032	75,168	69,600
April 2033	77,351	72,384
Completion: December 2033	77,351	77,351

Table 2: Number of Buses Implemented and Manufactured from January 2024 to December 2033 in Australia.

This table shows the complete schedule of manufacturing and implementing the e-buses in Australia. As it can be seen, the number of e-buses in the area gradually increased (696 buses). This means that there will be 77,256 e-buses to replace the 77,195 diesel buses - meeting the requirement of a full e-bus fleet.

Through this analysis of the e-bus fleet, there are two linear functions that can be utilized,

$$r(i) = 8352i$$

This function shows us the number of e-buses that will be manufactured during a given time. The variable i is the number of years that have passed starting 2024.

While

$$c(y) = 8352(y - 2024) + 2784$$

This function shows us the number of e-buses that are implemented at a given time starting from 2024. The variable y is the actual year.

With all of this in mind, this is the created roadmap for the transition to Australia.

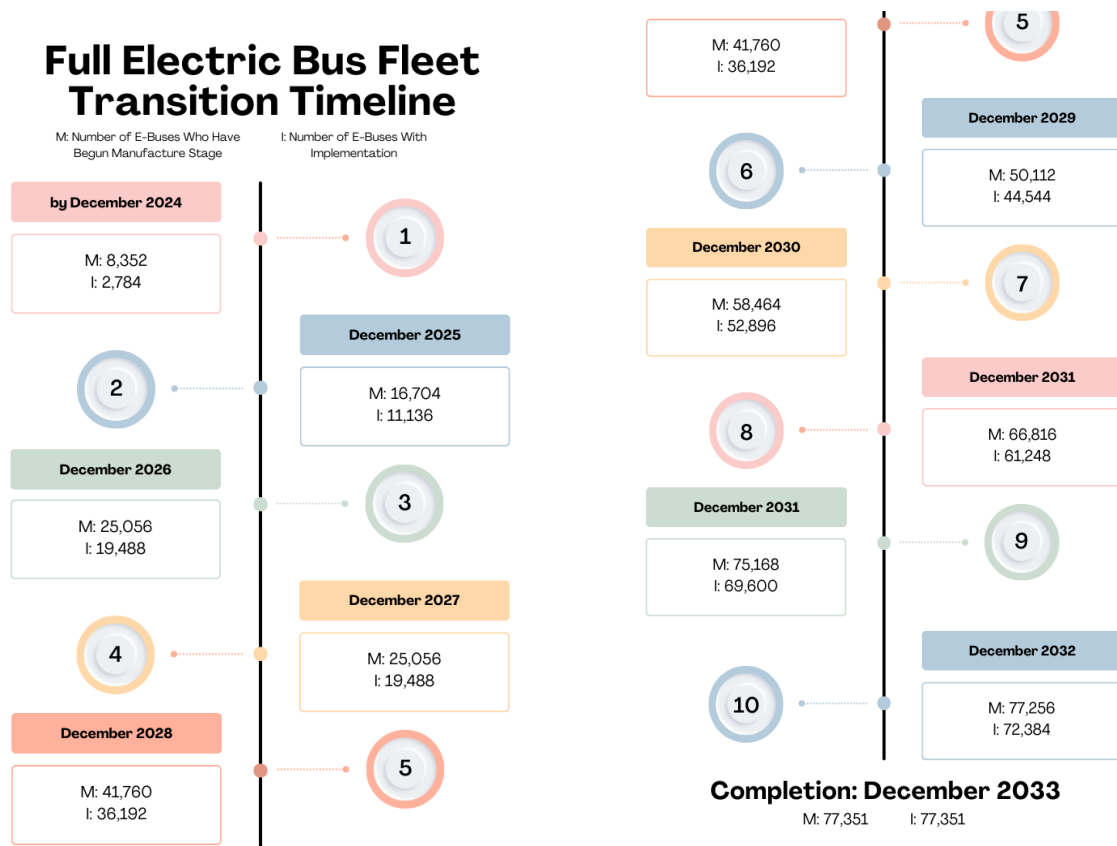


Image 2: Roadmap of Transition To a Full Electric Bus Fleet For Australia by 2033.

With this, the same analysis can also be applied to the other two metropolitan areas which are Tokyo, Japan, and Seoul, South Korea.

TOKYO, JAPAN	Number of E-Buses Who Have Begun Manufacture Stage	Number of E-Buses With Implementation
by December 2024	446	148
December 2025	891	594

December 2026	1,337	1,039
December 2027	1,782	1,485
December 2028	2,228	1,930
December 2029	2,673	2,376
December 2030	3,119	2,821
December 2031	3,568	3,267
December 2032	4,010	3,712
April 2033	4,121	3,861
Completion: December 2033	4,121	4,121

Table 3: Number of Buses Implemented and Manufactured from January 2024 to December 2033 in Tokyo, Japan.

SEOUL, SOUTH KOREA	Number of E-Buses Who Have Begun Manufacture Stage	Number of E-Buses With Implementation
by December 2024	5,460	1,820
December 2025	10,920	7,280
December 2026	16,380	12,740
December 2027	21,840	18,200
December 2028	27,300	23,660
December 2029	32,760	29,120
December 2030	38,220	34,580
December 2031	43,680	40,040
December 2032	49,140	45,500
April 2033	50,960	47,320
Completion: December 2033	50,960	50,960

Table 4: Number of Buses Implemented and Manufactured from January 2024 to December 2033 in Seoul, South Korea.

3 | Letter to Transportation Officials

13th of November 2023

Hopkins, Michael Ectric Bus
CEO of National Transport Commission
The University of Melbourne, Grattan Street
Parkville, Victoria, 3010, Australia

Good day!

Our Australian Bus and Coach Diesel Emissions (ABDCE) team hopes your much appreciated time spent in discussing the ongoing major e-bus project transitions into reduced complications for the program. You might have been informed that each state has begun their transition to replace the country's diesel buses to a fully electric fleet.

Firstly, we have identified and projected key financial expenses supported by electric buses that may improve the practicality of the transition. The draining costs are new technology and construction of the e-buses and their charging stations, the societal costs, and financial and ecological spendings due to less gas emissions, health costs, and noise pollution. Ecological benefits save us money, while initial construction is our main expense as well as the maintenance costs per month that will propagate as long as the e-bus is running.

The first part of our model took into account these factors and we have identified that based on the 20 billion budget of the "Rewiring the Nation" prime minister project, the more buses we make, the more ecological profit we earn which is the long-term larger profit. It will cost 24.2 billion dollars to replace every diesel bus with an electric bus, but ultimately 77,351 buses should be constructed as exceeding will result in no diesel buses to correspond to ecological profit. The second part is the roadmap to give an idea of what would happen in a 10-year time constraint, which is the construction of 696 buses every month over a 111-month period (about 9.25 years) with leading manufacturers such as Yutong Australia whose production limit is 20,000 electric buses every year. The money paid for transportation by passengers amounts to 12.7 billion over the 10-year period, which will support the missing funds needed.

May the 10-year roadmap and its root predictions serve as a precedent for each phase of the project, as the comparison between diesel bus and electric bus expenditure is the basis. Of course, we are open as always for issues in the details, and hope the time of your interest translates into the decided time of the complete transition.

Electrically,



ABDCE Team 14900

4 | Conclusion

Each of our models were able to show the effects of different restrictions or conditions on the number of buses constructed, the portion of the budget spent, and the ecological impact. The first condition maximizing the ecological profit showed that the best profit was achieved by spending all of the 20 billion budget in the projected time of 15 years. It takes 38,675 e-buses to start profiting ecologically (making it a long-term investment), and not by keeping a low total cost using C++. The assumption that up to over 7000 buses at a time could be in development, and finished e-buses may immediately replace diesel buses led to the second projection, the roadmap with the unique condition of a 10-year time constraint. The roadmap proved that the budget was large enough to accommodate the replacement of every diesel bus given that some of the money paid by e-bus passengers over the 10-year period will be used for funds to complete the missing 4.2 billion for construction.

A few limitations were the assumptions made in order to simplify the problem. This will lead to inaccuracies in the values, but the bare model can still be utilized to make it more accurate, while changing some variables to be not constant. Another limitation is that if a country does not have a lot of budget for a transition, a full transition can take about 500 years. As shown by the application of the model for financial implications, the most optimal way to transition is by using up the entire budget since that will take less time. Since the model is given through a set of equations, this means it is versatile and the values and constraints can be set as desired.

As the world progresses to a greener environment, transportation needs to be fixed as it is one of the leading causes of pollution. However, there are many complications such as the financial impacts it has, and the capabilities and resources of a country. The model provided predicts all of that, and offers a way for officials to decide their next step of action.

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